

Covariance vs Correlation (Complete Notes)

Introduction

In statistics and Machine Learning, Covariance and Correlation are two important measures used to understand the relationship between two variables.

For example:

Hours Studied → Exam Score

Height → Weight

Advertising Cost → Sales

Temperature → Ice Cream Sales

Both tell us whether two variables move together, but they differ in how they measure that relationship.

What is Covariance?

Definition

Covariance measures the direction of the relationship between two variables.

It tells us:

Do the variables increase together?

Does one increase while the other decreases?

Is there no relationship?

Formula

For a population:

$\text{Cov}(X, Y) =$

$\frac{1}{N}$

$\sum (X_i - \mu_X)(Y_i - \mu_Y)$

$- \mu_X$

X_i

$- \mu_Y$

Y_i

$- \mu_X$

X_i

$- \mu_Y$

Y_i

For a sample:

$$\text{Cov}(X,Y)=\frac{1}{n-1}\sum_{i=1}^n(X_i - \bar{X})(Y_i - \bar{Y})$$

$$X_i - \bar{X}$$

$$Y_i - \bar{Y}$$

Where:

X = First variable

Y = Second variable

$\bar{X}$

= Mean of X

$\bar{Y}$

= Mean of Y

Interpretation of Covariance

Positive Covariance

$$\text{Cov}(X,Y)>0$$

Both variables move in the same direction.

Example:

Study Hours	Marks
2	50
4	60
6	75
8	90

As study hours increase, marks increase.

Positive covariance.

Negative Covariance

$$\text{Cov}(X,Y)<0$$

Variables move in opposite directions.

Example:

TV Watching	Marks
8	50
6	60
4	75
2	90

More TV → Lower marks.

Negative covariance.

Zero Covariance

$\text{Cov}(X,Y)=0$

No linear relationship exists.

Example:

Shoe Size	IQ
6	110
7	95
8	130
9	105

No meaningful relationship.

Advantages of Covariance

Easy to compute

Shows direction of relationship

Foundation of covariance matrices in ML

Useful in finance (portfolio analysis)

Disadvantages of Covariance

The biggest limitation:

It does not tell the strength of the relationship.

Example:

Dataset A:

Covariance = 120

Dataset B:

Covariance = 5

Can we say Dataset A has a stronger relationship?

No.

Why?

Because covariance depends on the units of measurement.

If we measure height in centimeters instead of meters, covariance changes.

Therefore, covariance values cannot be compared across datasets.

What is Correlation?

Correlation is a standardized version of covariance.

It measures:

Direction
Strength

of a linear relationship.

Unlike covariance, correlation always lies between -1 and +1.

Formula

$r =$

$\frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}$

σ_X
 σ_Y

$\text{Cov}(X,Y)$

Where:

$\text{Cov}(X,Y)$ = Covariance

σ_X
 σ_Y

= Standard deviation of X

σ_X
 σ_Y

= Standard deviation of Y

Because covariance is divided by the standard deviations, the units cancel out, making correlation unitless.

Range of Correlation

$-1 \leq r \leq +1$

Perfect Positive Correlation

$r = +1$

Variables move exactly together.

Example:

X : 1 2 3 4

Y : 2 4 6 8

Every increase in X produces a proportional increase in Y.

Strong Positive Correlation

$r = 0.9$

Almost perfect positive relationship.

Moderate Positive

$r = 0.6$

Positive relationship with some variation.

Weak Positive

$r = 0.2$

Very little positive relationship.

No Correlation

$r=0$

No linear relationship.

Example:

Height

Movie Rating

Weak Negative

$r=-0.2$

Slight inverse relationship.

Strong Negative

$r=-0.9$

Variables strongly move in opposite directions.

Perfect Negative

$r=-1$

Example:

X : 1 2 3 4

Y : 8 6 4 2

As X increases, Y decreases perfectly.

Visualizing Correlation

Drag points

2

4

6

8

10

5

10

X
Y

Linear correlation is positive ($r = 0.98$).

Negative

Near zero

Positive

Negative

Near zero

Positive

Numerical Example

Suppose:

Hours	Marks
2	40
4	50
6	65
8	80

Mean Hours

5

Mean Marks

58.75

Compute deviations:

Hours Dev	Marks Dev
-3	-18.75
-1	-8.75
1	6.25
3	21.25

Multiply deviations:

56.25

8.75

6.25

63.75

Sum:

135

Sample covariance:

3

135

=45

Suppose

$\text{Std}(X)=2.58$

Std(Y)=17.5

Then

$r = \frac{2.58 \times 17.5}{45}$

$r \approx 0.996$

Very strong positive correlation.

Key Differences

Feature	Covariance	Correlation
Meaning	Direction of relationship	Direction and strength
Range	$-\infty$ to $+\infty$	-1 to +1
Units	Depends on units	Unitless
Comparison across datasets		Not meaningful Meaningful
Strength measurement	No	Yes
Standardized	No	Yes
Affected by scale	Yes	No
Relationship Between Them		

Correlation is simply the normalized form of covariance:

$$\text{Correlation} = \frac{\text{Std}(X) \times \text{Std}(Y)}{\text{Covariance}}$$

Covariance tells us which direction variables move, while correlation tells us which direction and how strongly they move.

Applications in Machine Learning

1. Feature Selection

Highly correlated features often provide redundant information.

Example:

Age
Years Since Birth

Keeping both may not improve the model.

2. Multicollinearity Detection

In regression models, highly correlated independent variables can lead to unstable coefficient estimates and reduced interpretability.

3. Data Analysis

Understanding relationships between variables before training a model.

4. Financial Modeling

Covariance and correlation help analyze how stocks move together for portfolio diversification.

5. Principal Component Analysis (PCA)

PCA uses the covariance matrix (or correlation matrix after standardization) to identify directions of maximum variance and reduce dimensionality.

Python Example

```
import pandas as pd
```

```
data = {  
    "Hours": [2, 4, 6, 8],  
    "Marks": [40, 50, 65, 80]  
}
```

```
df = pd.DataFrame(data)
```

```
print(df.cov()) # Covariance Matrix  
print(df.corr()) # Correlation Matrix
```

Output:

Covariance

	Hours	Marks
Hours	6.67	45.00
Marks	45.00	308.33

	Hours	Marks
Hours	1.00	0.996
Marks	0.996	1.00

Common Interview Questions

Can covariance be greater than 1?

Yes. Covariance has no fixed range.

Can correlation be greater than 1?

No. Correlation is always between -1 and $+1$.

Why is correlation preferred over covariance?

Because it is standardized, unitless, and allows meaningful comparisons across datasets.

Does zero correlation imply independence?

Not necessarily. Zero correlation only indicates no linear relationship. Variables may still have a nonlinear relationship.

What does a correlation of -0.85 mean?

It indicates a strong negative linear relationship: as one variable increases, the other tends to decrease.